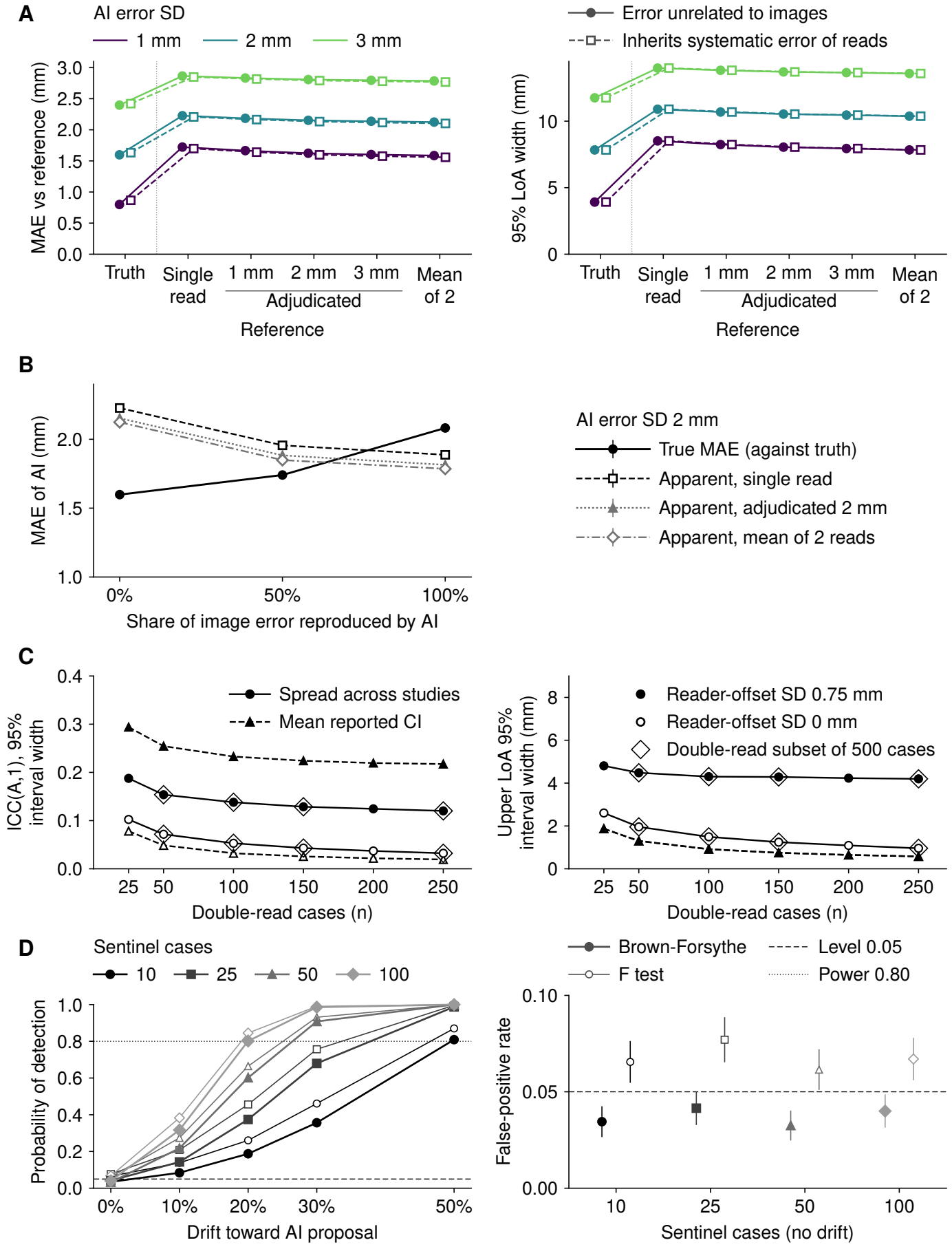
A
AI error SD
1 mm
2 mm
3 mm
MAE vs reference (mm)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Truth
Single read
1 mm
2 mm
3 mm
Adjudicated
Mean of 2
Reference
Error unrelated to images
Inherits systematic error of reads
95% LoA width (mm)
10
5
0
B
MAE of AI (mm)
2.0
1.5
1.0
0%
50%
100%
Share of image error reproduced by AI
AI error SD 2 mm
True MAE (against truth)
Apparent, single read
Apparent, adjudicated 2 mm
Apparent, mean of 2 reads
C
ICC(A,1), 95% interval width
0.4
0.3
0.2
0.1
0.0
Spread across studies
Mean reported CI
25
50
100
150
200
250
Double-read cases (n)
Upper LoA 95% interval width (mm)
8
6
4
2
0
Reader-offset SD 0.75 mm
Reader-offset SD 0 mm
Double-read subset of 500 cases
D
Sentinel cases
10
25
50
100
Probability of detection
1.0
0.8
0.6
0.4
0.2
0.0
0%
10%
20%
30%
50%
Drift toward AI proposal
Brown-Forsythe
F test
Level 0.05
Power 0.80
False-positive rate
0.10
0.05
0.00
Sentinel cases (no drift)